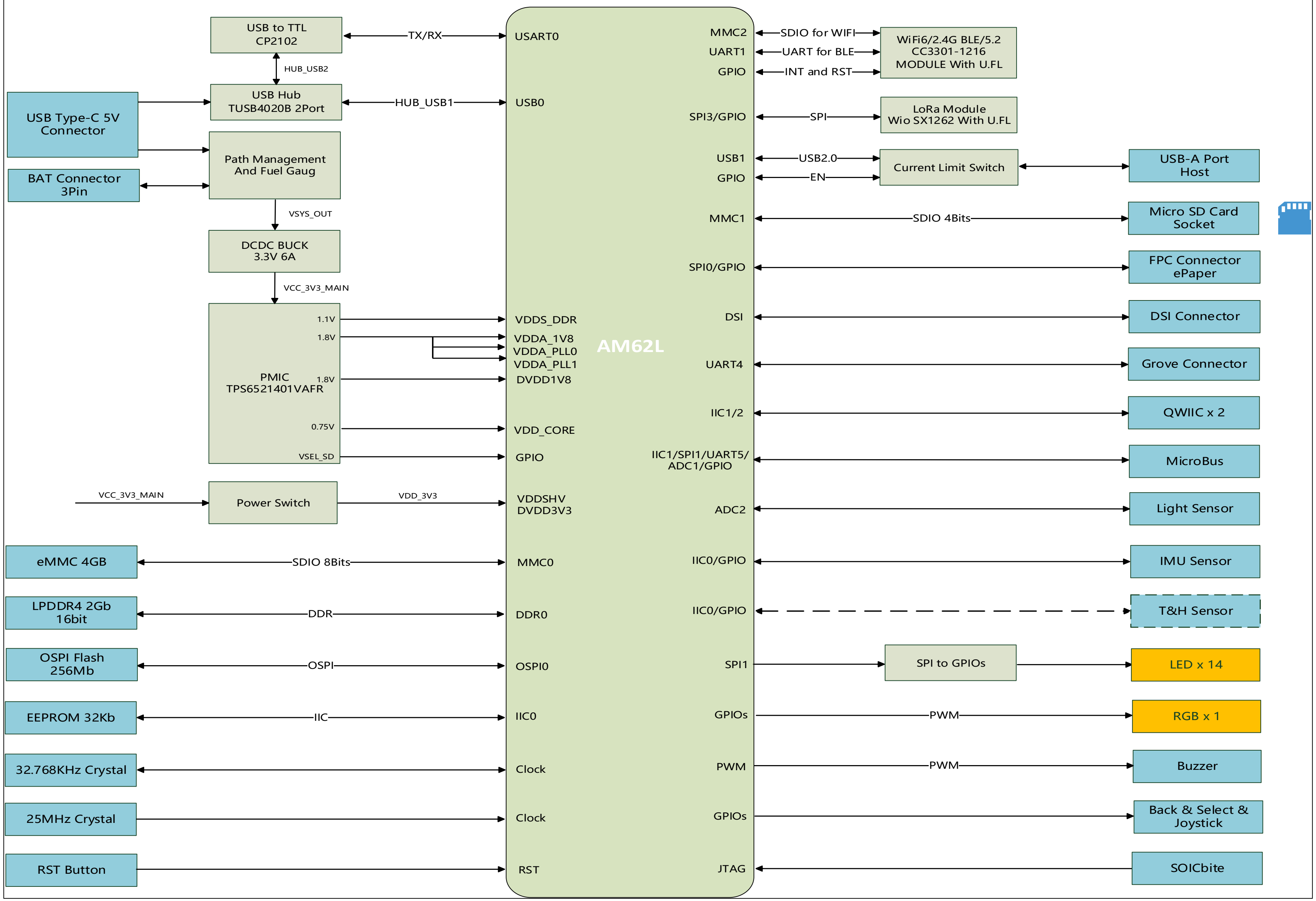
Schematic:BeagleBadge AM62L

Revision History

DATE	REVISION	DESCRIPTION
2025/10/14	Rev A	1. Initial
2025/10/20	V0.3	Main modifications: 1. Change the capacitor withstand voltage to 2x or higher. 2. Correct the schematic symbol of the AND gate. 3. Add switch control to the power monitoring ADC. 4. Modify the circuit of RTC_PG and PMIC to reset AM62L. 5. DNI the output capacitor of TPS22965DSGR. 6. Add eMMC circuit. 7. Correct the wake-up signal(WAKEUP0/1). 8. Change Tag-Connect to SOIC16 Bite. 9. Change button gpios to 3V3 IOs because connect to SDIOBite 10. DNI I2C singles unpopulated for compatibility touchscreens.
2025/10/24	V0.4	1. Replace U9 with 0R resistors to switch Low Power mode. 2. Boot and Select function share one button. 3. Replace the Up/Down/Left/Right buttons with a joystick 4. Use the SPI1 to connect U42 (SPI to GPIO) and mikroBUS 5. Correct and add network identification descriptions 6. Adjust some GPIO allocation.
2025/10/28	V0.5	1. Added power supply selection for the IMU. 2. Modified the U62 circuit to prevent its input from floating. 3. Added the EXT_WAKEUP0 signal to pin 12 of J29 (SOICbite).
2025/10/29	V0.6	1. Use M23/GPIO0_43/3V3 to reset the USB Hub. 2. Use F22/GPIO0_29/PWM/3V3 to drive the buzzer. 3. The RGB LED has been changed to a smaller package.
2025/11/05	V0.7	1. Page8: Correct the VDDS0 power supply voltage to 1V8. 2. Page19: The JTAG signals change to 3.3V through a buffer. 3. Page24: The "Other GPIOs" pins shall be changed to 1V8.

Block Diagram



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Size:	Document Number:	Rev:
A3	02 Block Diagram	RevA
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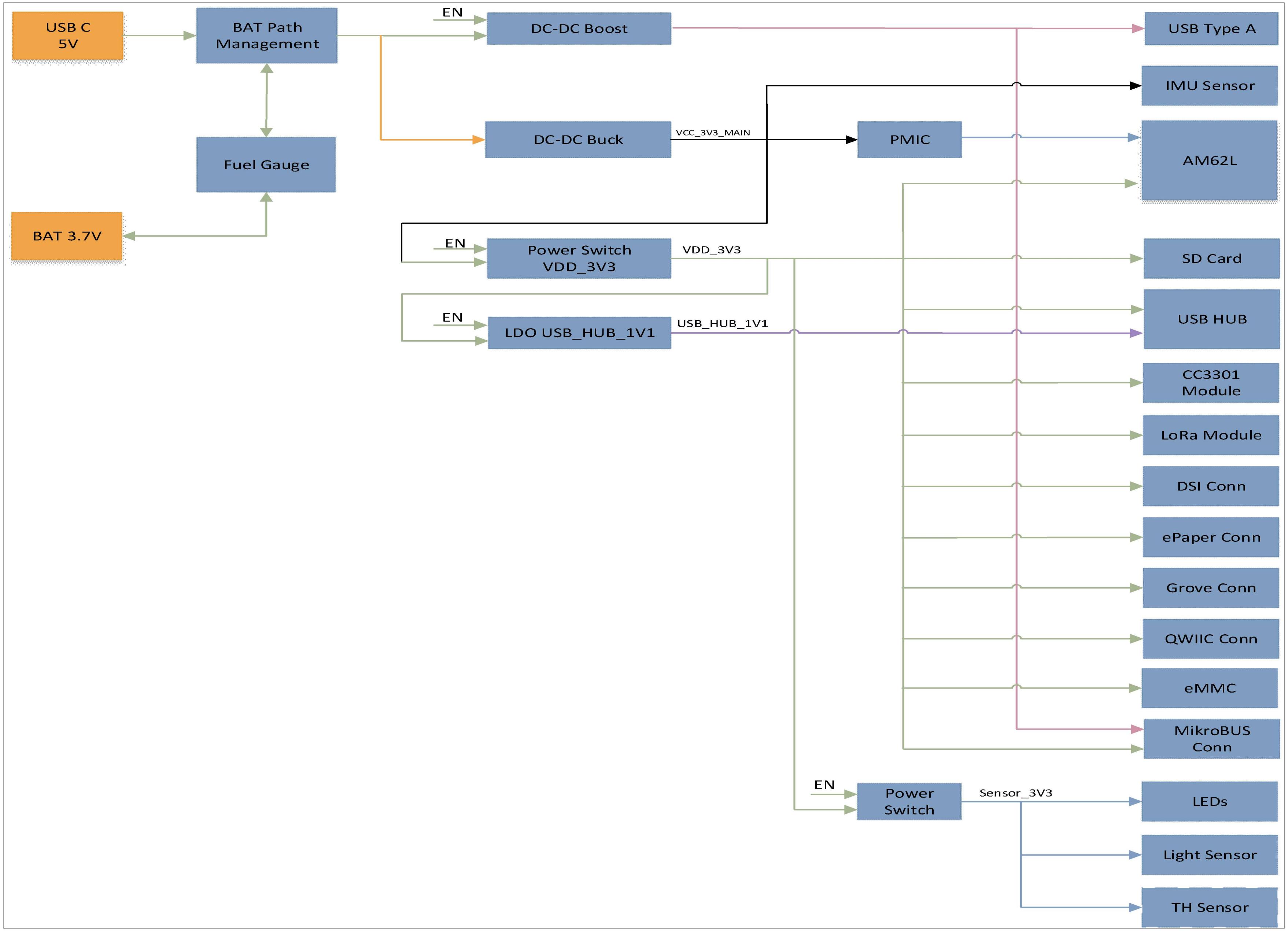

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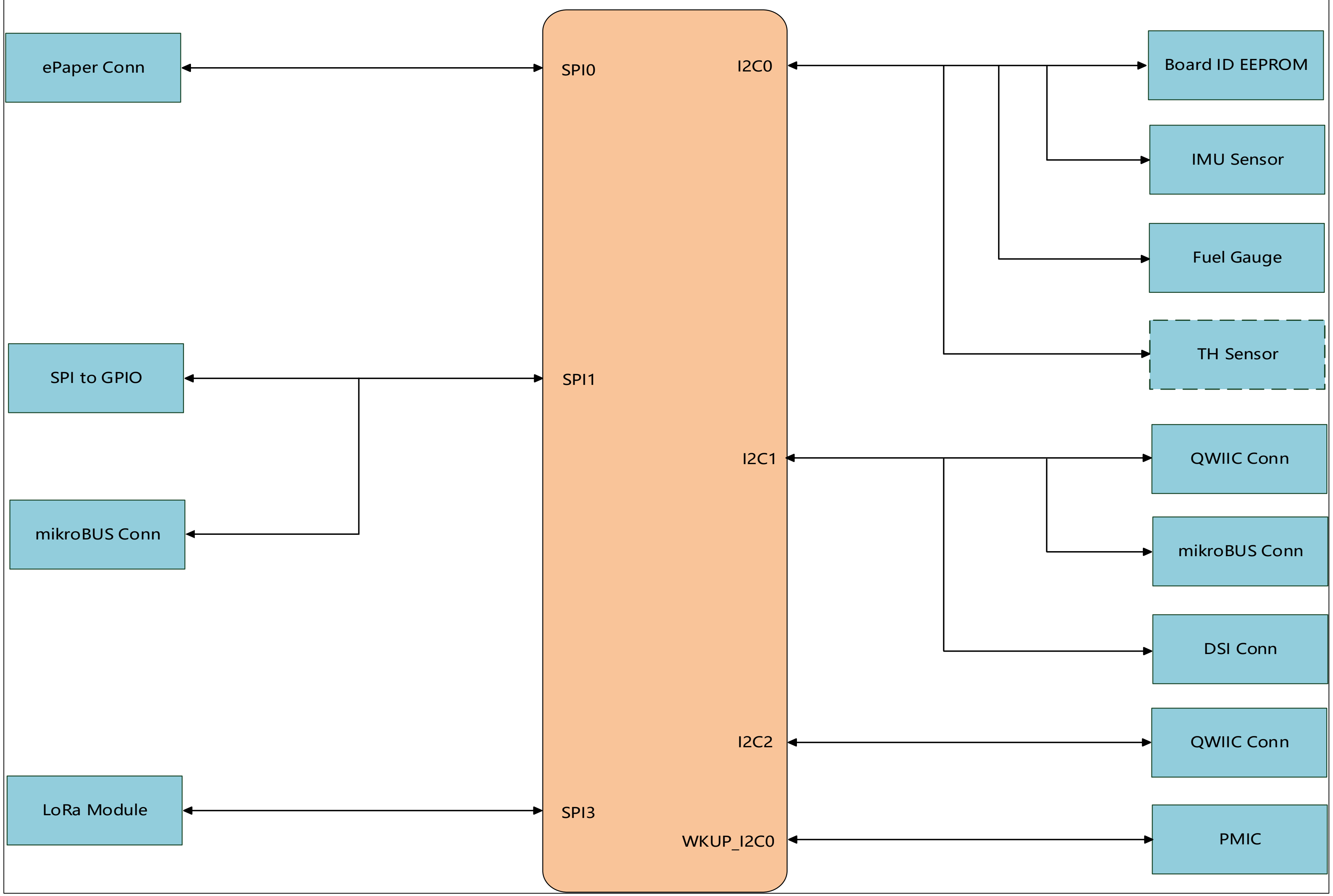
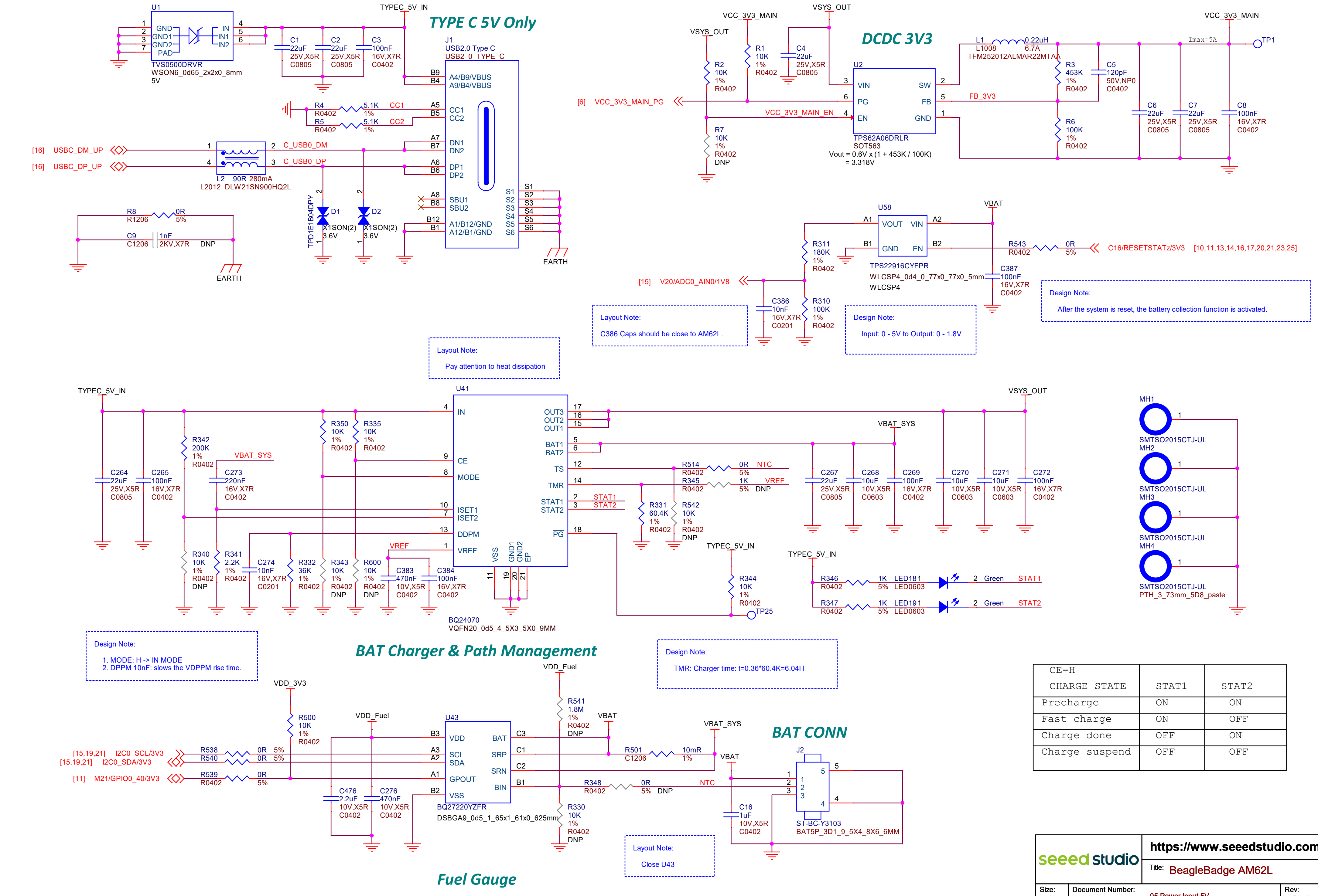
SPI & I2C TREE


ePaper Conn
SPI0
I2C0
Board ID EEPROM
IMU Sensor
Fuel Gauge
TH Sensor
SPI to GPIO
SPI1
mikroBUS Conn
I2C1
QWIIC Conn
mikroBUS Conn
DSI Conn
I2C2
QWIIC Conn
LoRa Module
SPI3
WKUP_I2C0
PMIC



CE=H CHARGE STATE	STAT1	STAT2
Precharge	ON	ON
Fast charge	ON	OFF
Charge done	OFF	ON
Charge suspend	OFF	OFF

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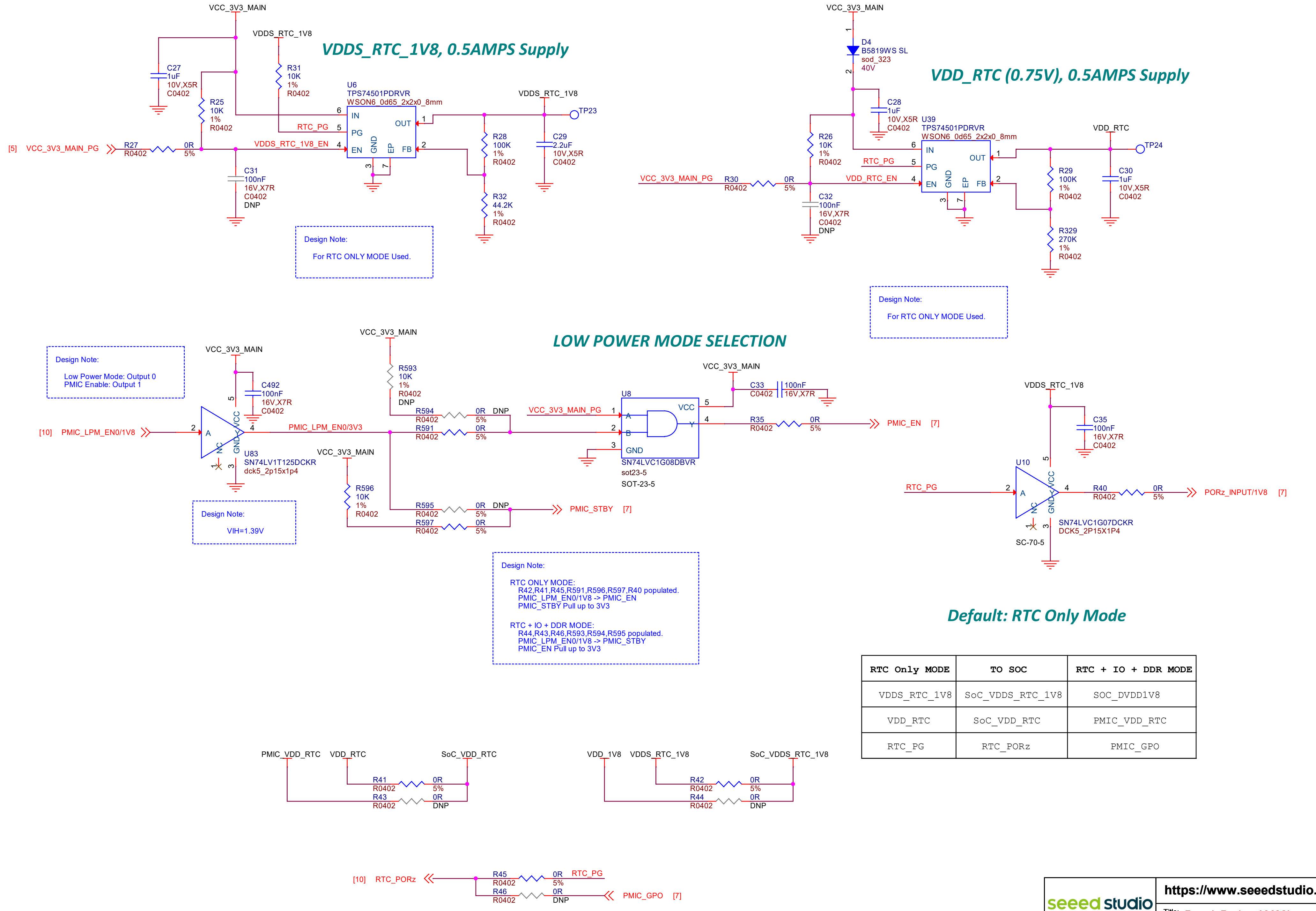

VDDS_RTC_1V8, 0.5AMPS Supply

Design Note:
For RTC ONLY MODE Used.

VDD_RTC (0.75V), 0.5AMPS Supply

Design Note:
For RTC ONLY MODE Used.

LOW POWER MODE SELECTION

Design Note:
Low Power Mode: Output 0
PMIC Enable: Output 1

Design Note:
VIH=1.39V

Design Note:

RTC ONLY MODE:
R42,R41,R45,R591,R596,R597,R40 populated.
PMIC_LPM_EN0/1V8 -> PMIC_EN
PMIC_STBY Pull up to 3V3

RTC + IO + DDR MODE:
R44,R43,R46,R593,R594,R595 populated.
PMIC_LPM_EN0/1V8 -> PMIC_STBY
PMIC_EN Pull up to 3V3

Default: RTC Only Mode

RTC Only MODE	TO SOC	RTC + IO + DDR MODE
VDDS_RTC_1V8	SoC_VDDS_RTC_1V8	SOC_DVDD1V8
VDD_RTC	SoC_VDD_RTC	PMIC_VDD_RTC
RTC_PG	RTC_PORz	PMIC_GPO

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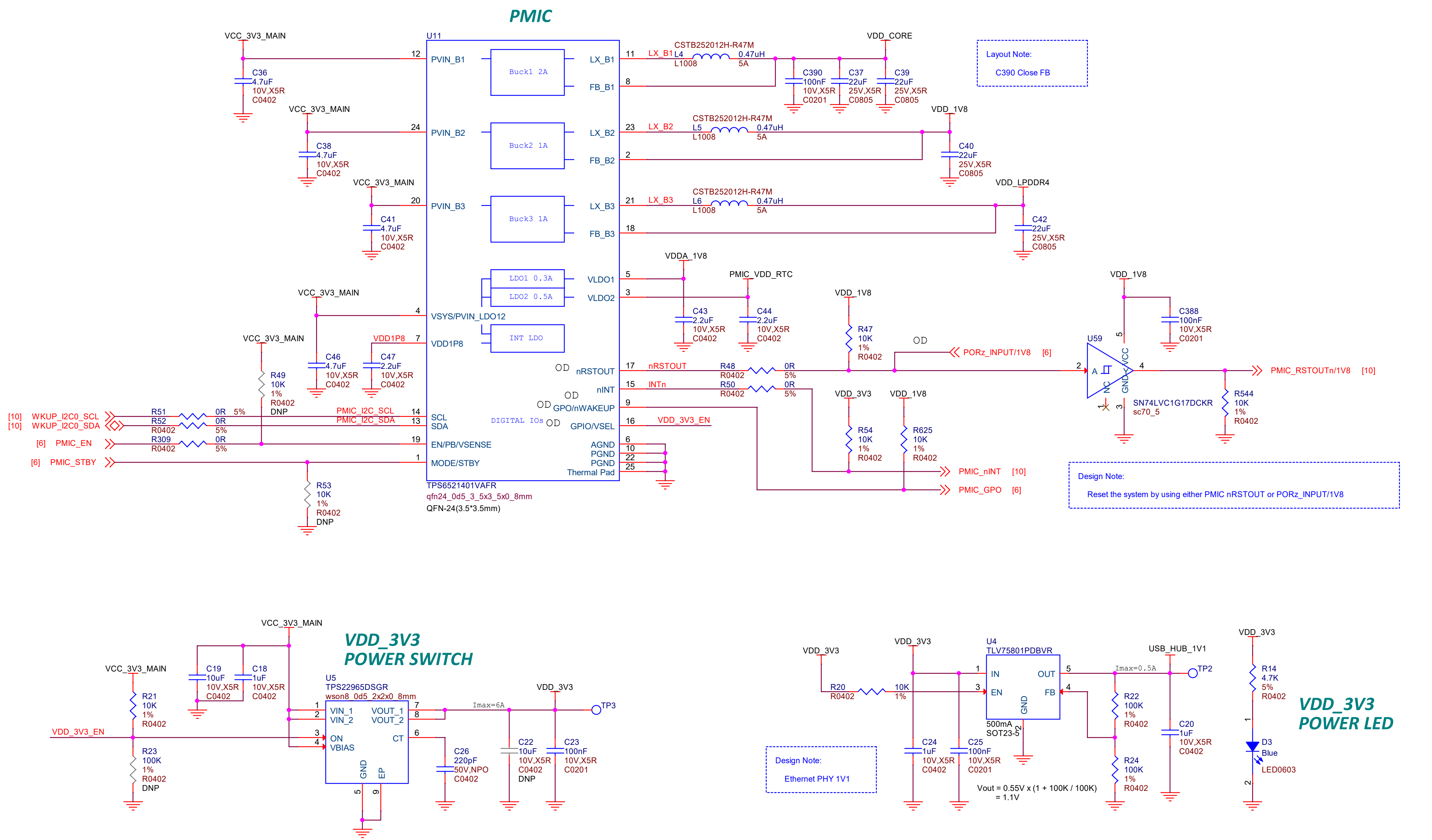
Size: A3
Document Number: 06 RTC ONLY Mode
Rev: RevA
Date: Friday, November 07, 2025



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Size:	Document Number:	Rev:
A3	07 AM62L PMIC	RevA

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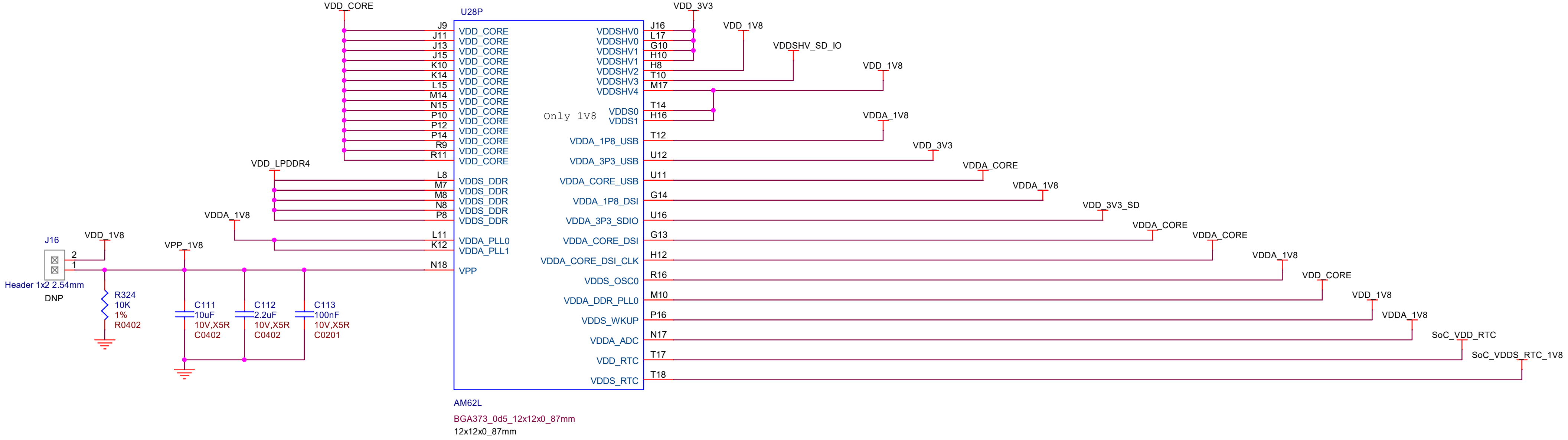
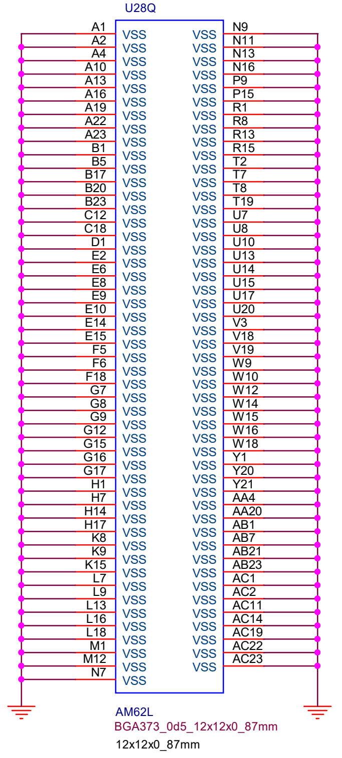
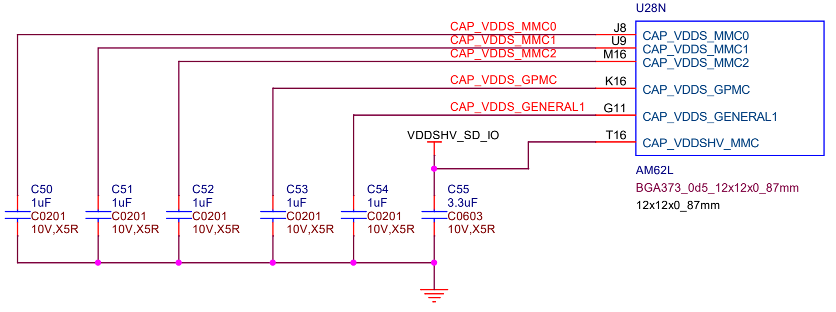
SOC POWER SUPPLIES AND SUPPLY RAILS



SOC VSS





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Title: BeagleBadge AM62L

Size: A3	Document Number: 08 AM62L Power	Rev: RevA
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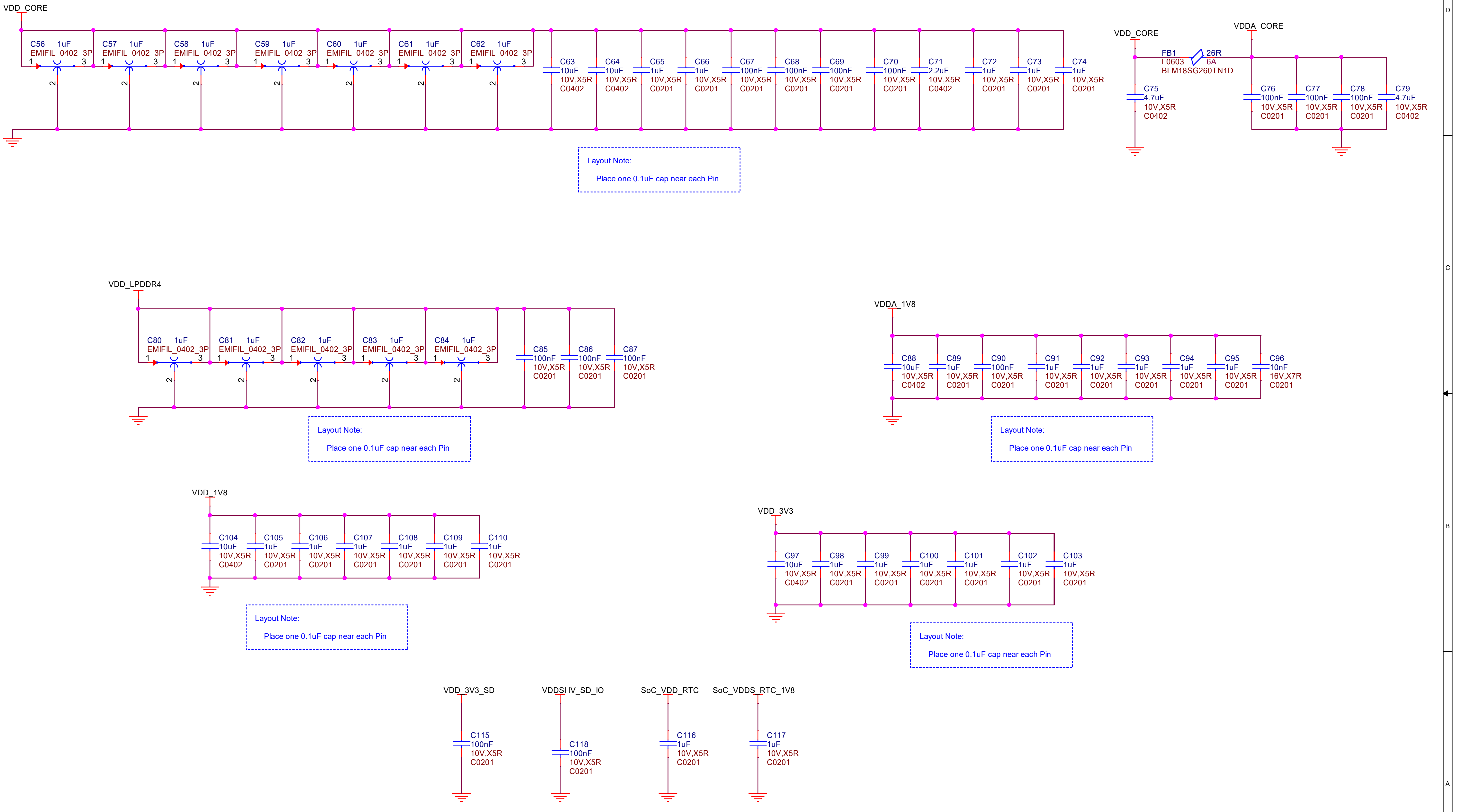
SOC POWER SUPPLIES - DECAPS



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Title: BeagleBadge AM62L

Size:	Document Number:	Rev:
A3	09 AM62L DCAPs	RevA

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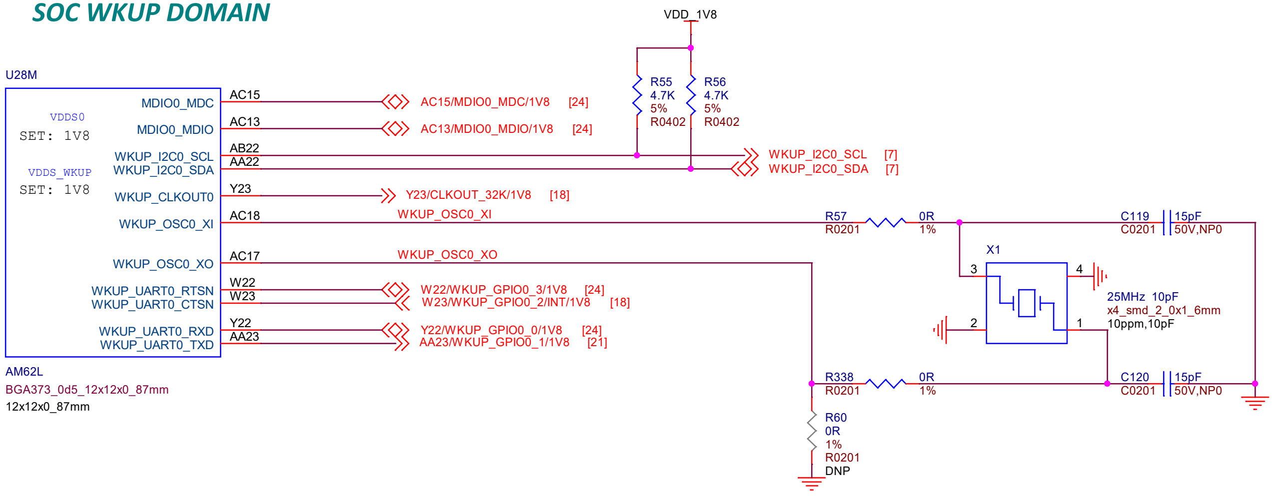
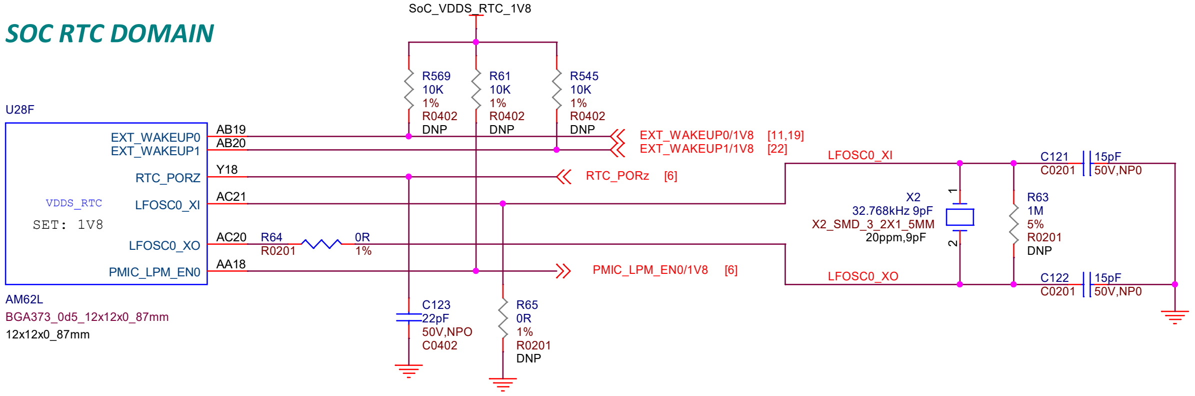
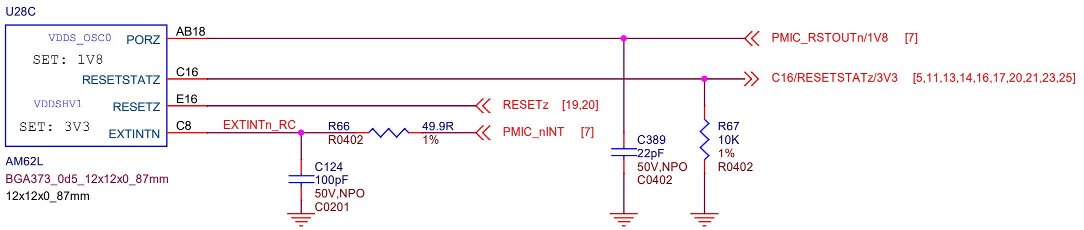
SOC WKUP DOMAIN



SOC RTC DOMAIN



SOC RESET



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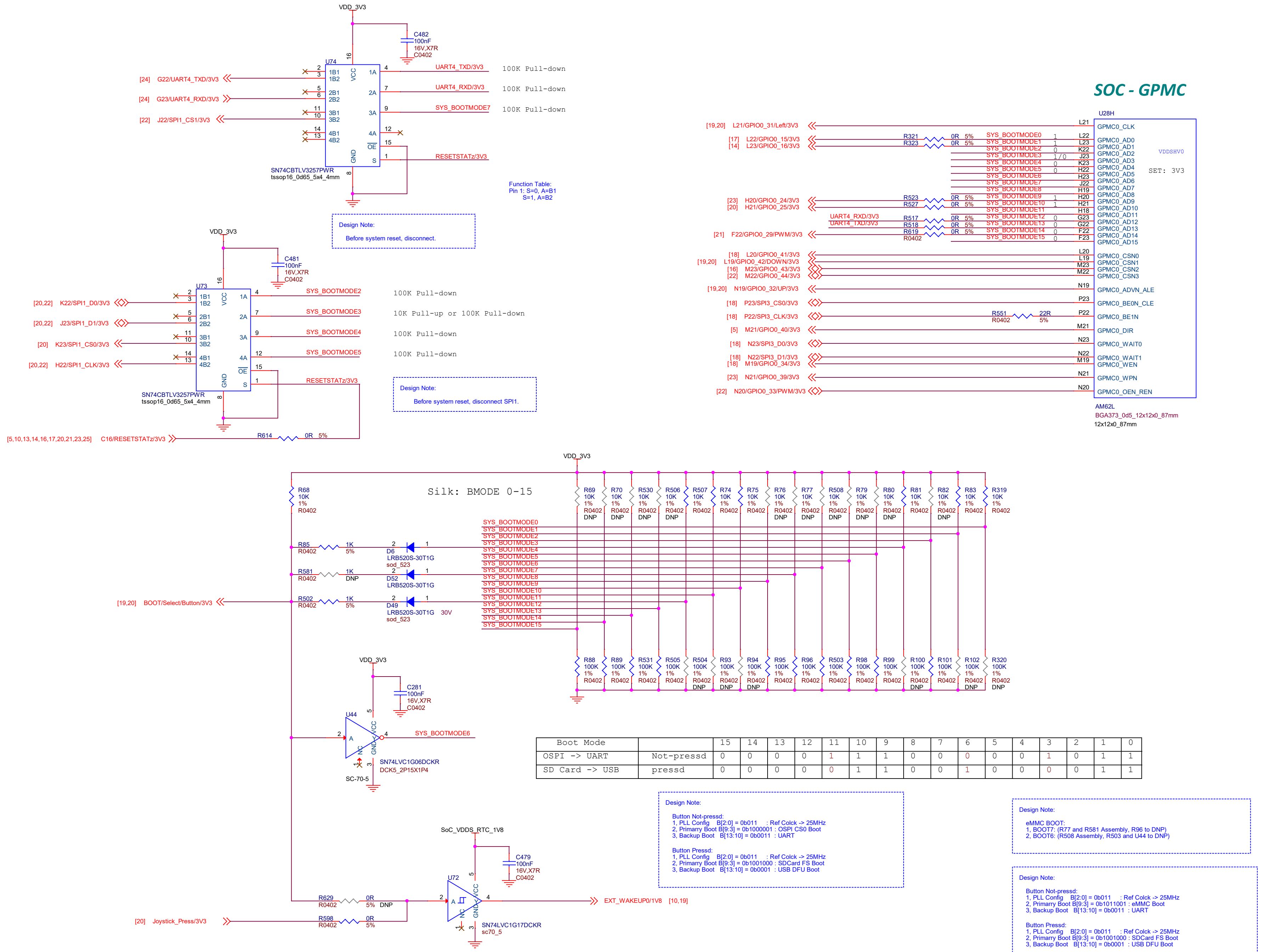
	Title: BeagleBadge AM62L
Size: A3	Document Number: 10 AM62L WKUP & OSC & Reset — Rev: RevA
Date: Friday, November 07, 2025	Sheet: 10 of 25



Boot Mode		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OSPI -> UART	Not-pressd	0	0	0	0	1	1	1	0	0	0	0	0	1	0	1	1
SD Card -> USB	pressd	0	0	0	0	0	1	1	0	0	1	0	0	0	0	1	1

Design Note:

Button Not-pressd:
1, PLL Config B[2:0] = 0b011 : Ref Colck -> 25MHz
2, Primarry Boot B[9:3] = 0b1000001 : OSPI CS0 Boot
3, Backup Boot B[13:10] = 0b0011 : UART

Button Pressd:
1, PLL Config B[2:0] = 0b011 : Ref Colck -> 25MHz
2, Primarry Boot B[9:3] = 0b1001000 : SDCard FS Boot
3, Backup Boot B[13:10] = 0b0001 : USB DFU Boot

Design Note:

eMMC BOOT:
1, BOOT7: (R77 and R581 Assembly, R96 to DNP)
2, BOOT6: (R508 Assembly, R503 and U44 to DNP)

Design Note:

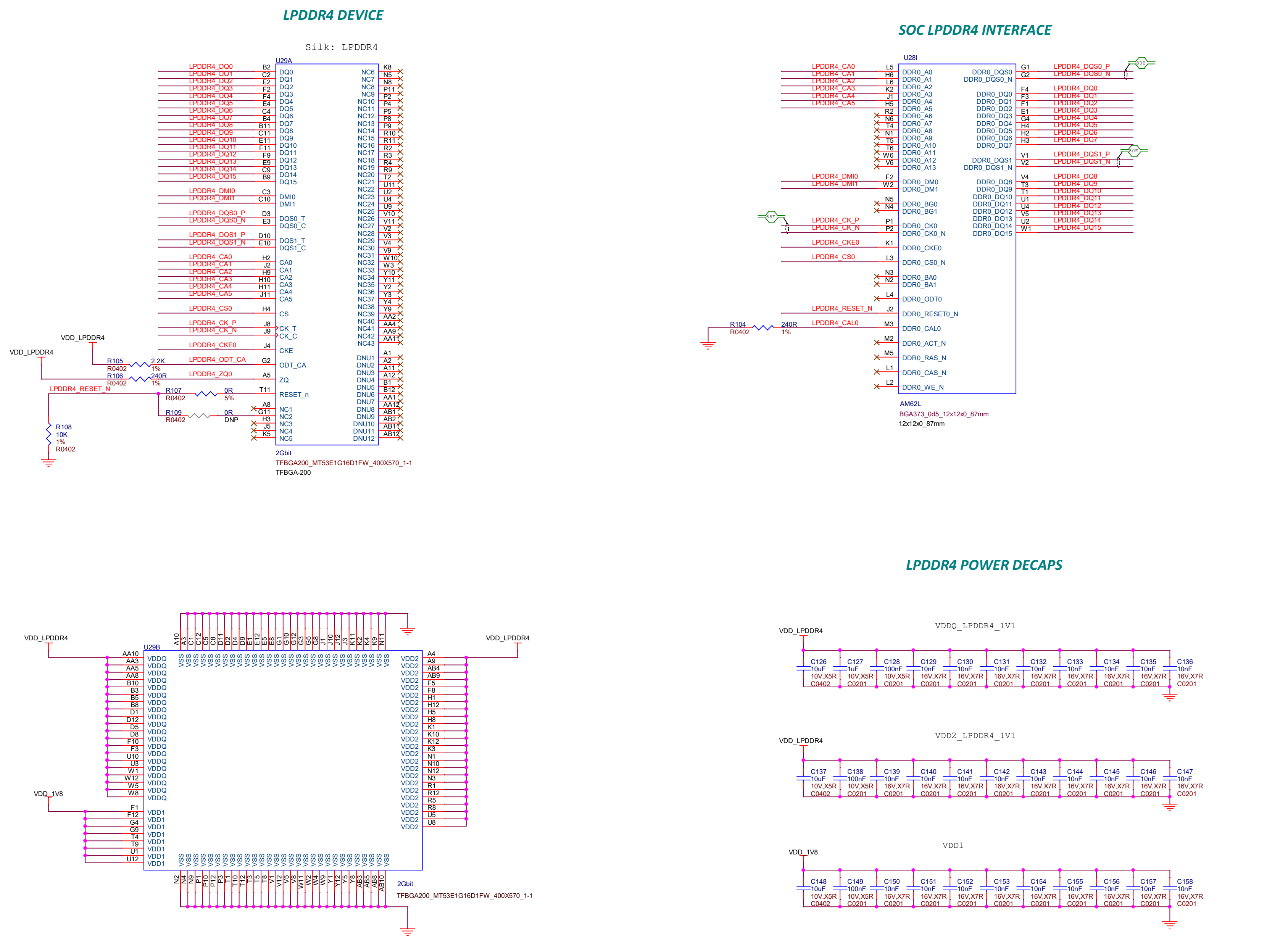
Button Not-pressd:
1, PLL Config B[2:0] = 0b011 : Ref Colck -> 25MHz
2, Primarry Boot B[9:3] = 0b1011001 : eMMC Boot
3, Backup Boot B[13:10] = 0b0011 : UART

Button Pressd:
1, PLL Config B[2:0] = 0b011 : Ref Colck -> 25MHz
2, Primarry Boot B[9:3] = 0b1001000 : SDCard FS Boot
3, Backup Boot B[13:10] = 0b0001 : USB DFU Boot

Boot Mode		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
eMMC -> UART	Not-pressd	0	0	0	0	1	1	1	0	1	1	0	0	1	0	1	1
SD Card -> USB	pressd	0	0	0	0	0	1	1	0	0	1	0	0	0	0	1	1

Design Note:

Use the Joystick_Press Button or Select Button to wake-up System.



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Title: BeagleBadge AM62L		
Size: A2	Document Number: 12 AM62L LPDDR4 DEVICE	Rev: RevA
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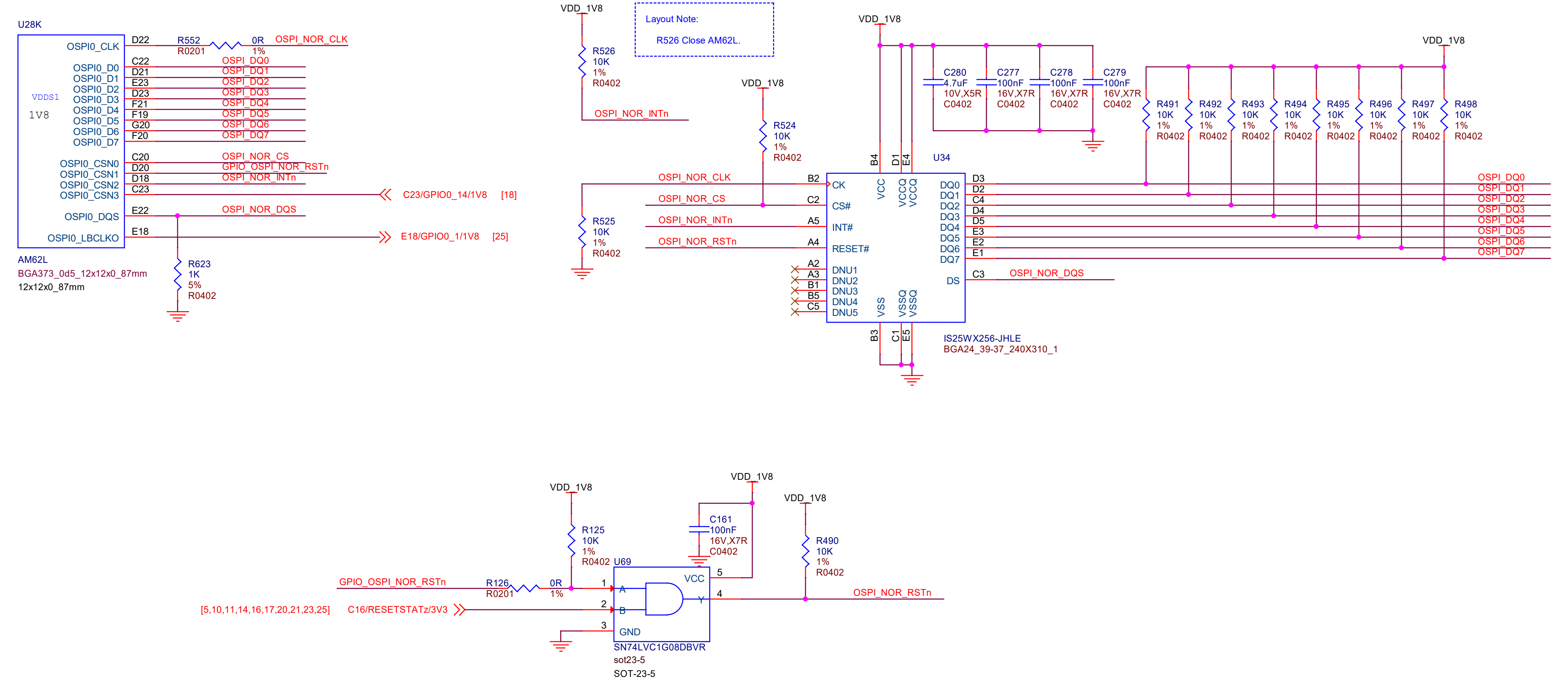
SOC OSPI INTERFACE

OSPI NOR FLASH



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Title: BeagleBadge AM62L		
Size: A3	Document Number: 13 AM62L OSPI Flash	Rev: RevA
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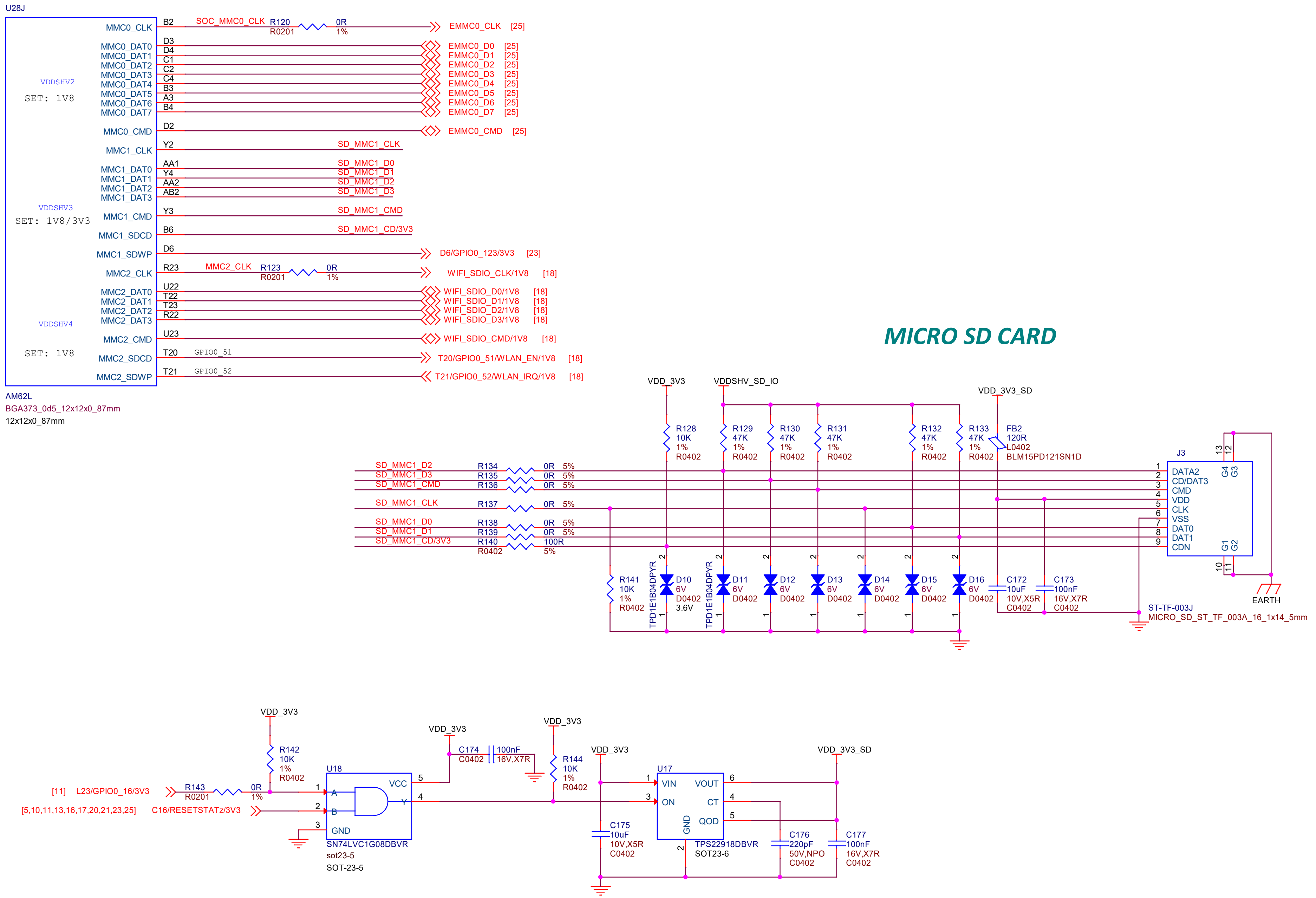
SOC - MMC Interface



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	Title: BeagleBadge AM62L
Size: A3	Document Number: 14 AM62L Micro SD — Rev: RevA
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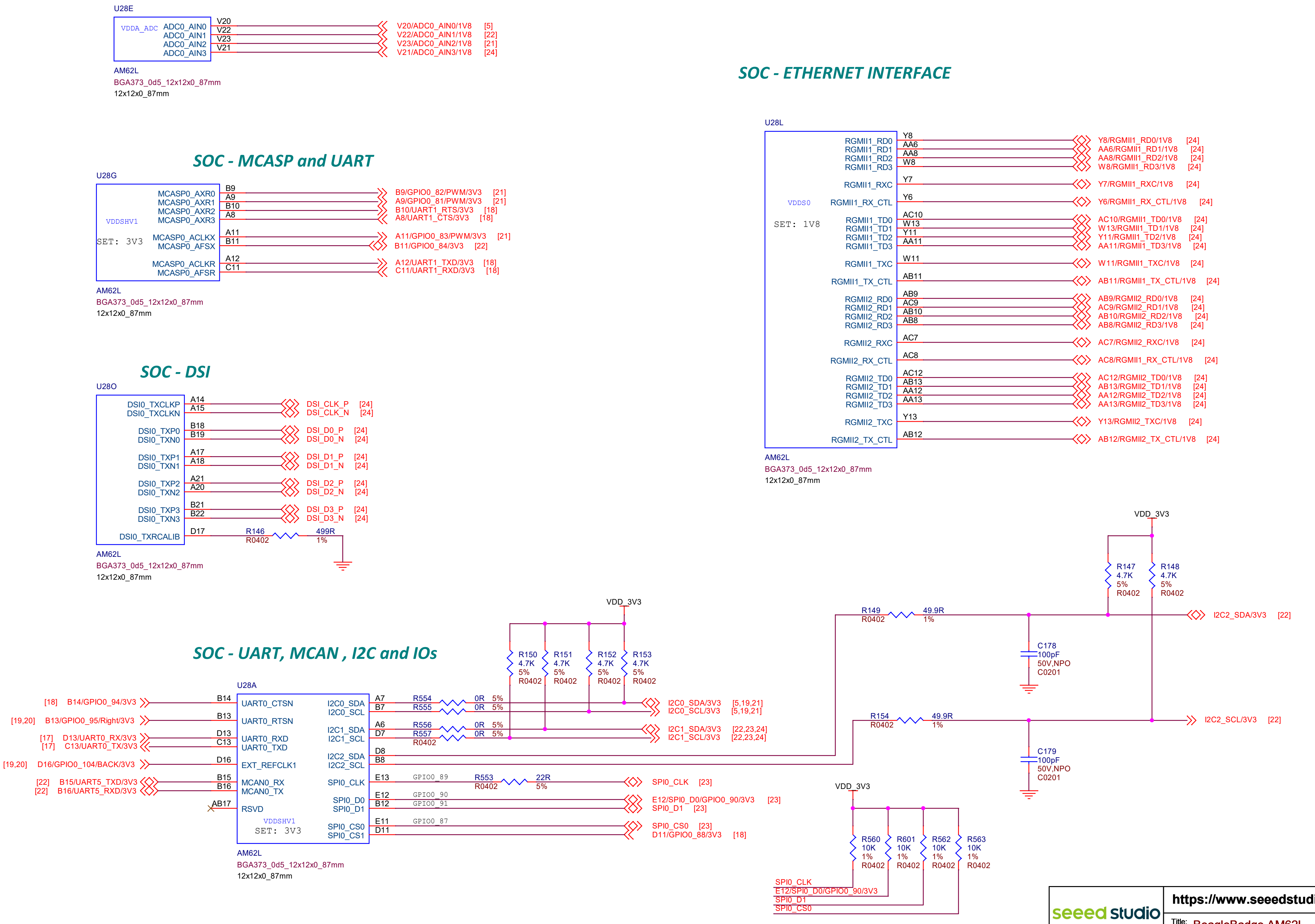

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Title: BeagleBadge AM62L

Size: A3	Document Number: 15 AM62L ADC & DSI & OTHER	Rev: RevA
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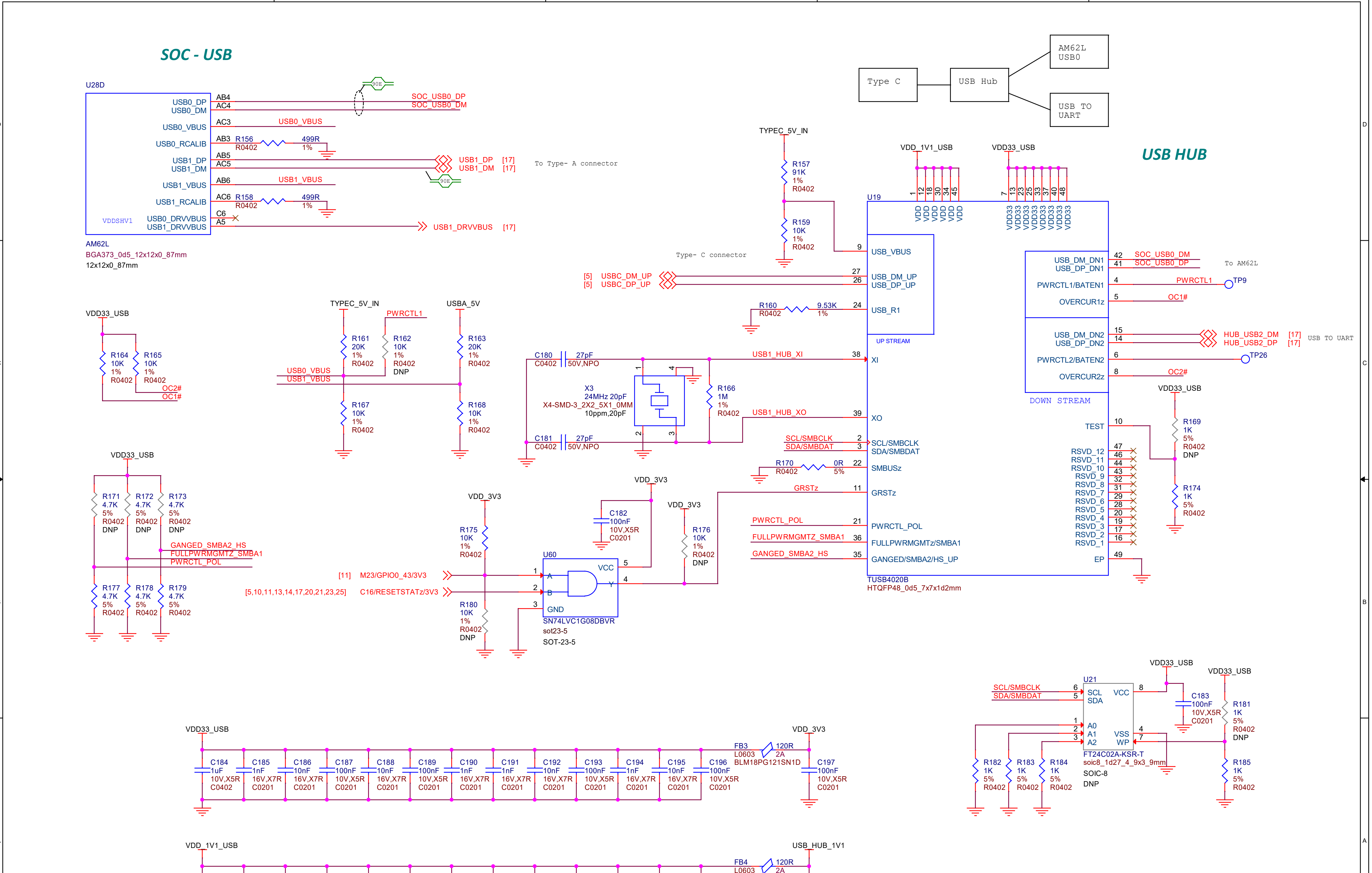

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Title: BeagleBadge AM62L

Size: A3	Document Number: 16 AM62L USB & HUB	Rev: RevA
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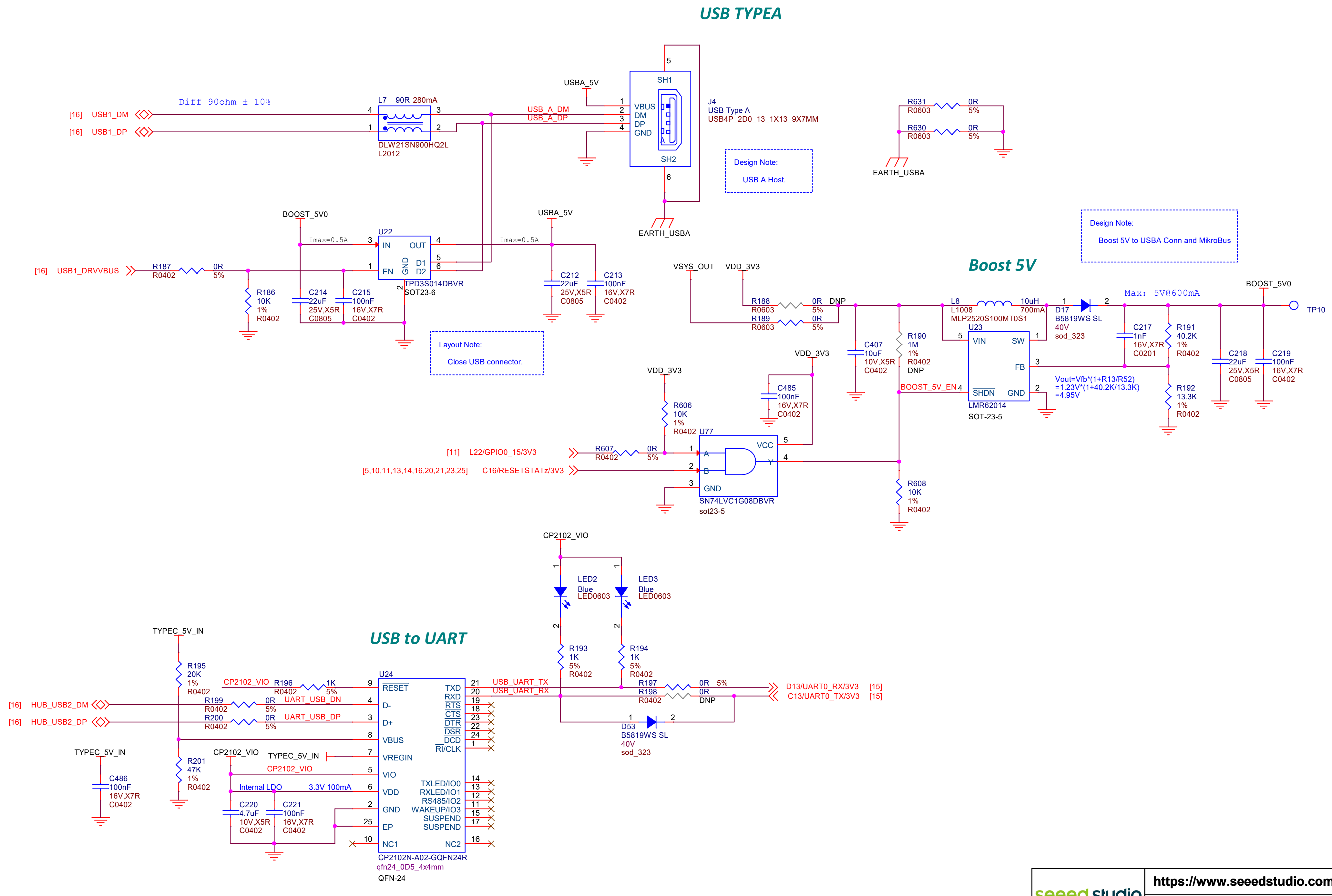

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Title: BeagleBadge AM62L		
Size: A3	Document Number: 17 USB2.0 Type A Host & UART	Rev: RevA
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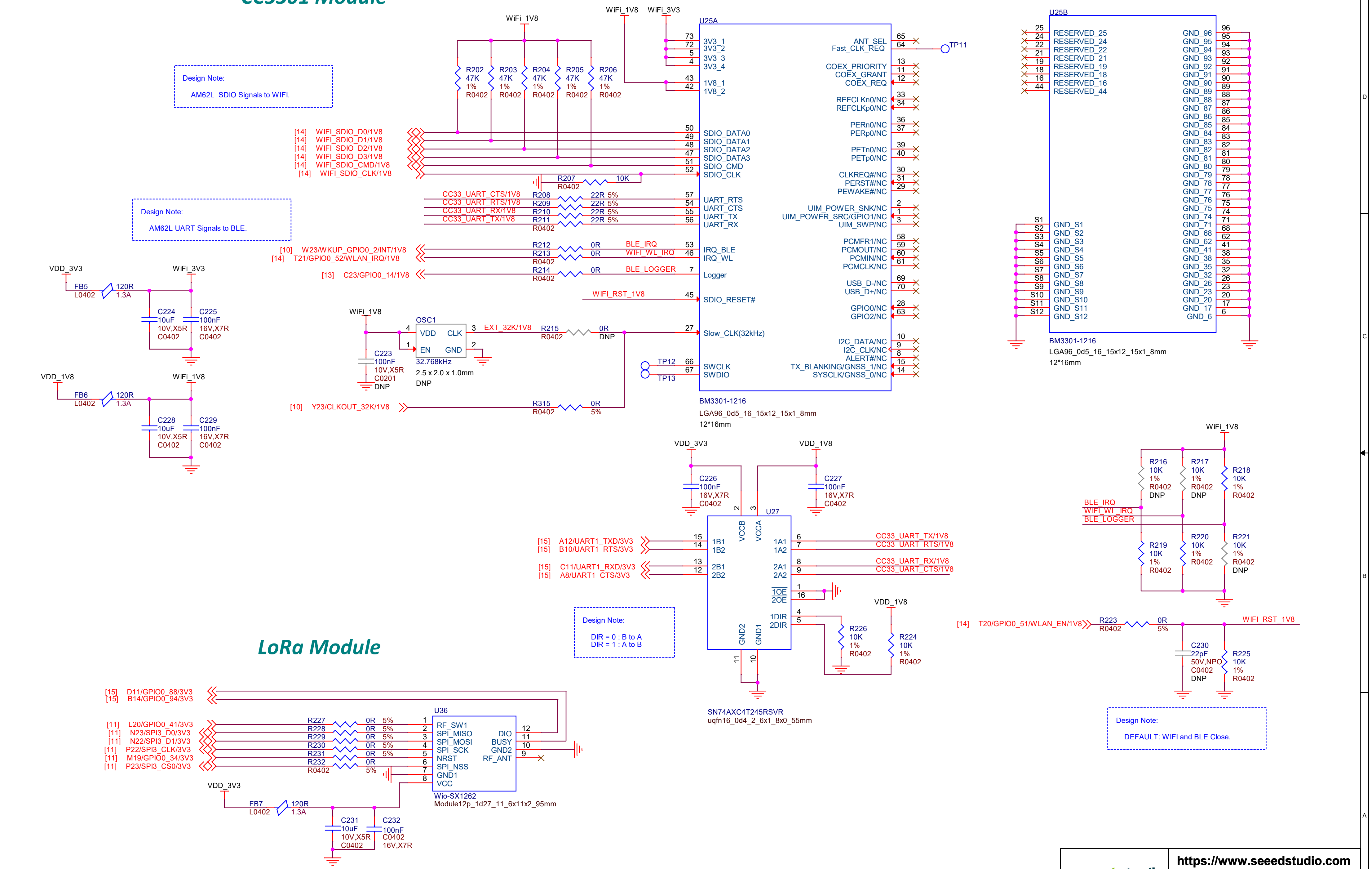
CC3301 Module



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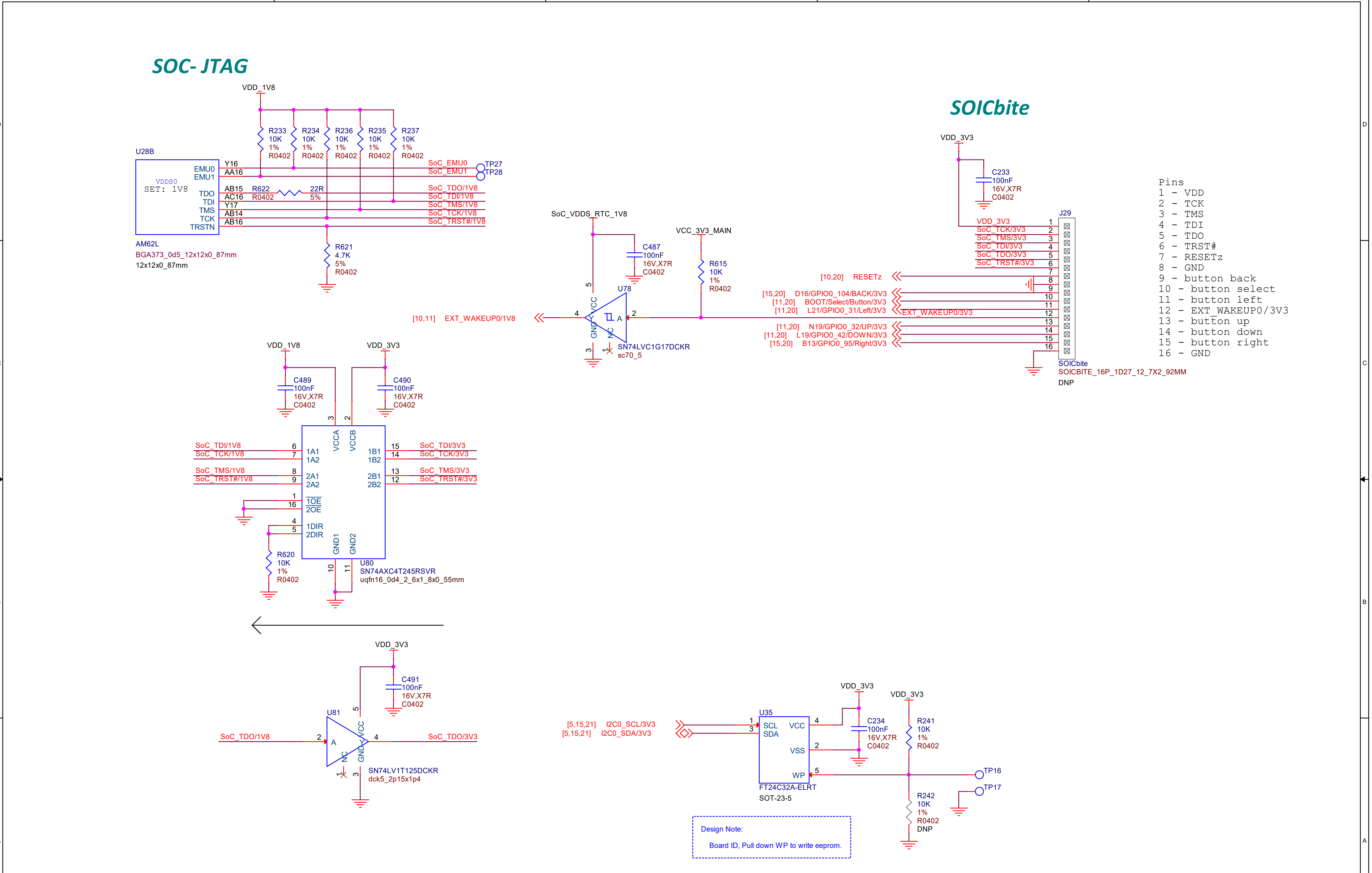
	Title: BeagleBadge AM62L	
Size: A3	Document Number: 18 BM3301 & Sx1262 Module	Rev: RevA
Date: Friday, November 07, 2025		Sheet: 18 of 25



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Title: BeagleBadge AM62L

Size: A3 | Document Number: 19 EEPROM & SOICbite | Rev: RevA

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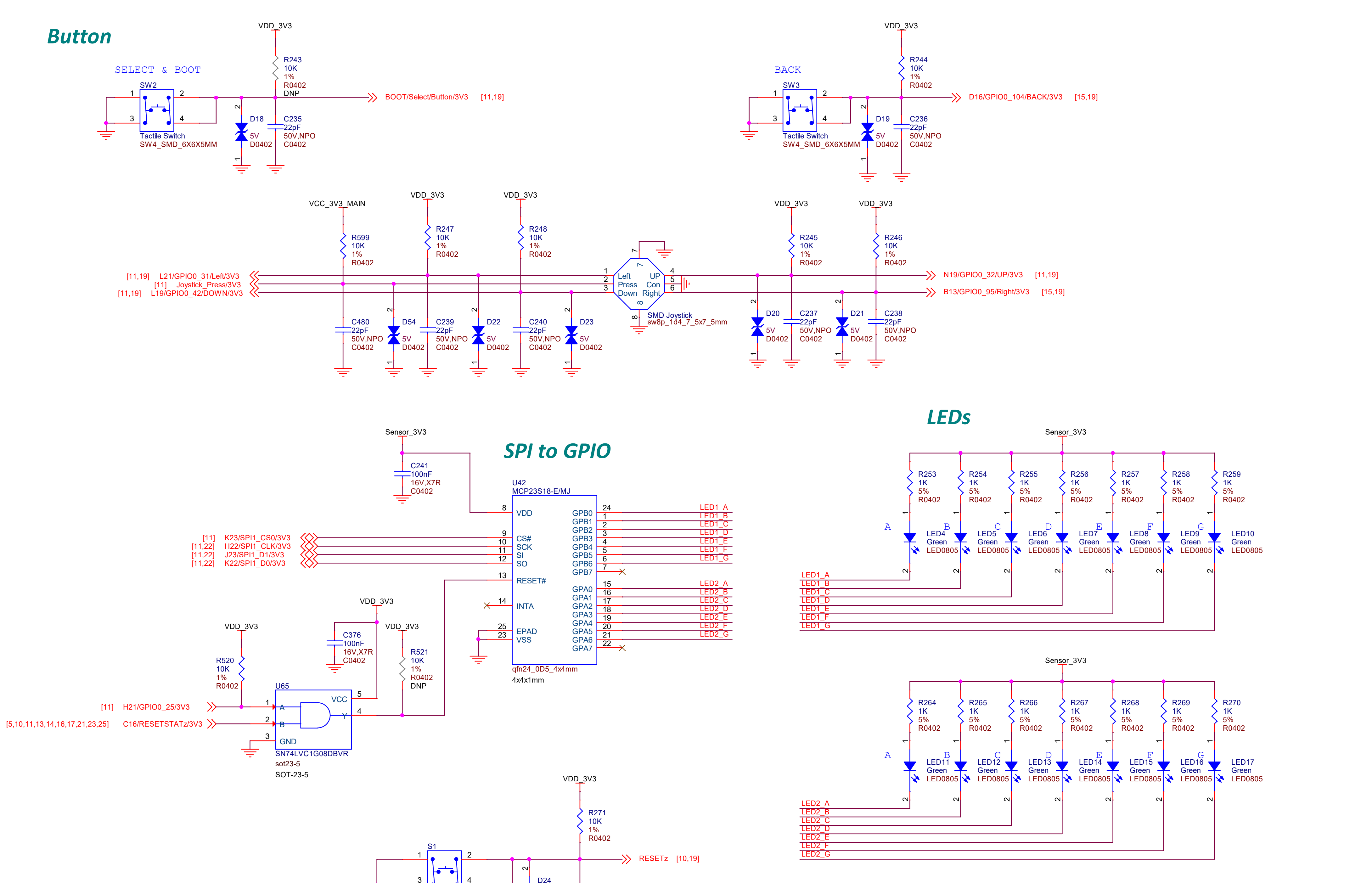

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Title: BeagleBadge AM62L			
Size: A3	Document Number: 20 Button & LEDs		Rev: RevA
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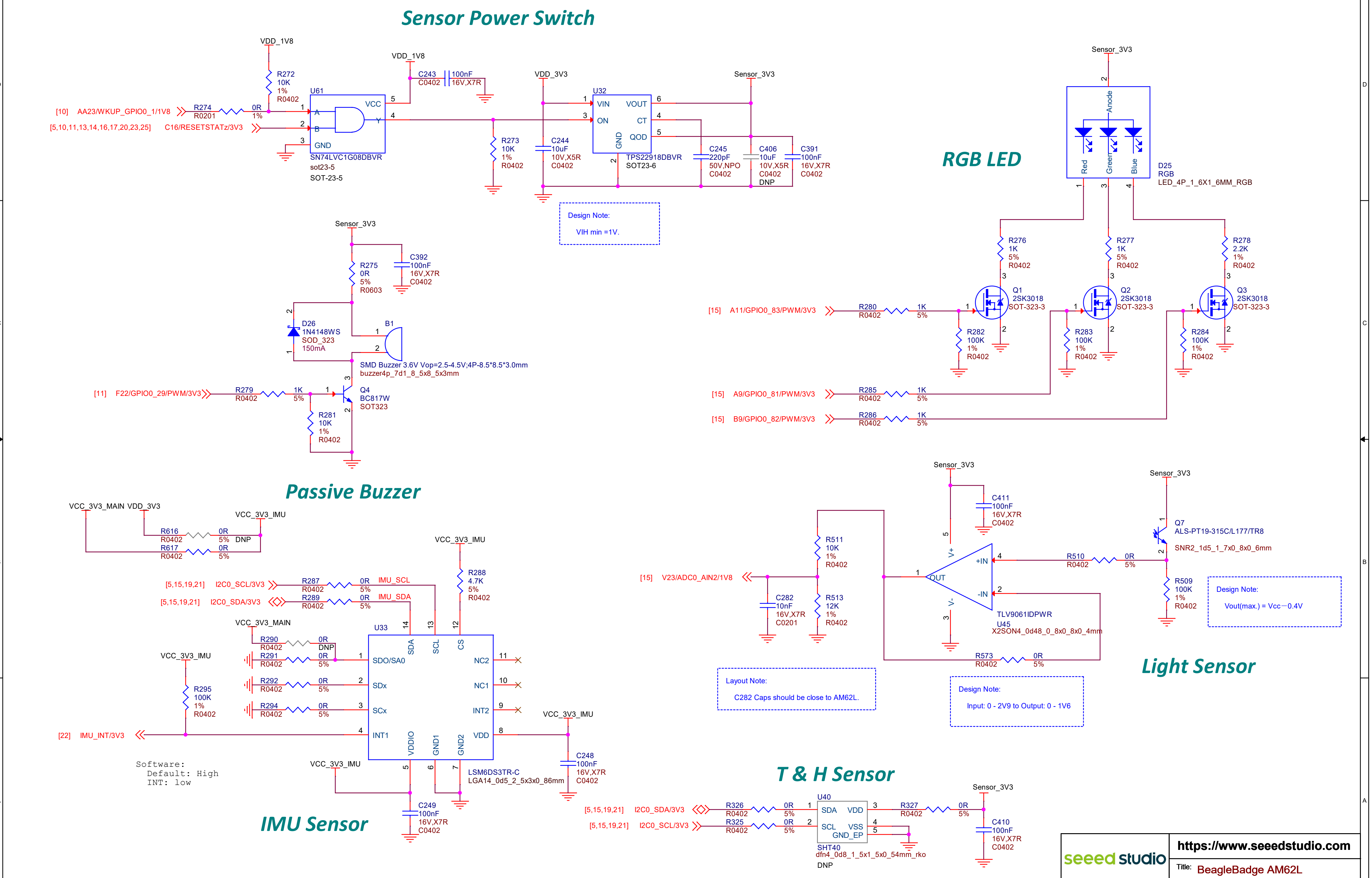

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Title:	BeagleBadge AM62L
Size: A3	Document Number: 21 Buzzer & IMU & Light
Rev:	RevA
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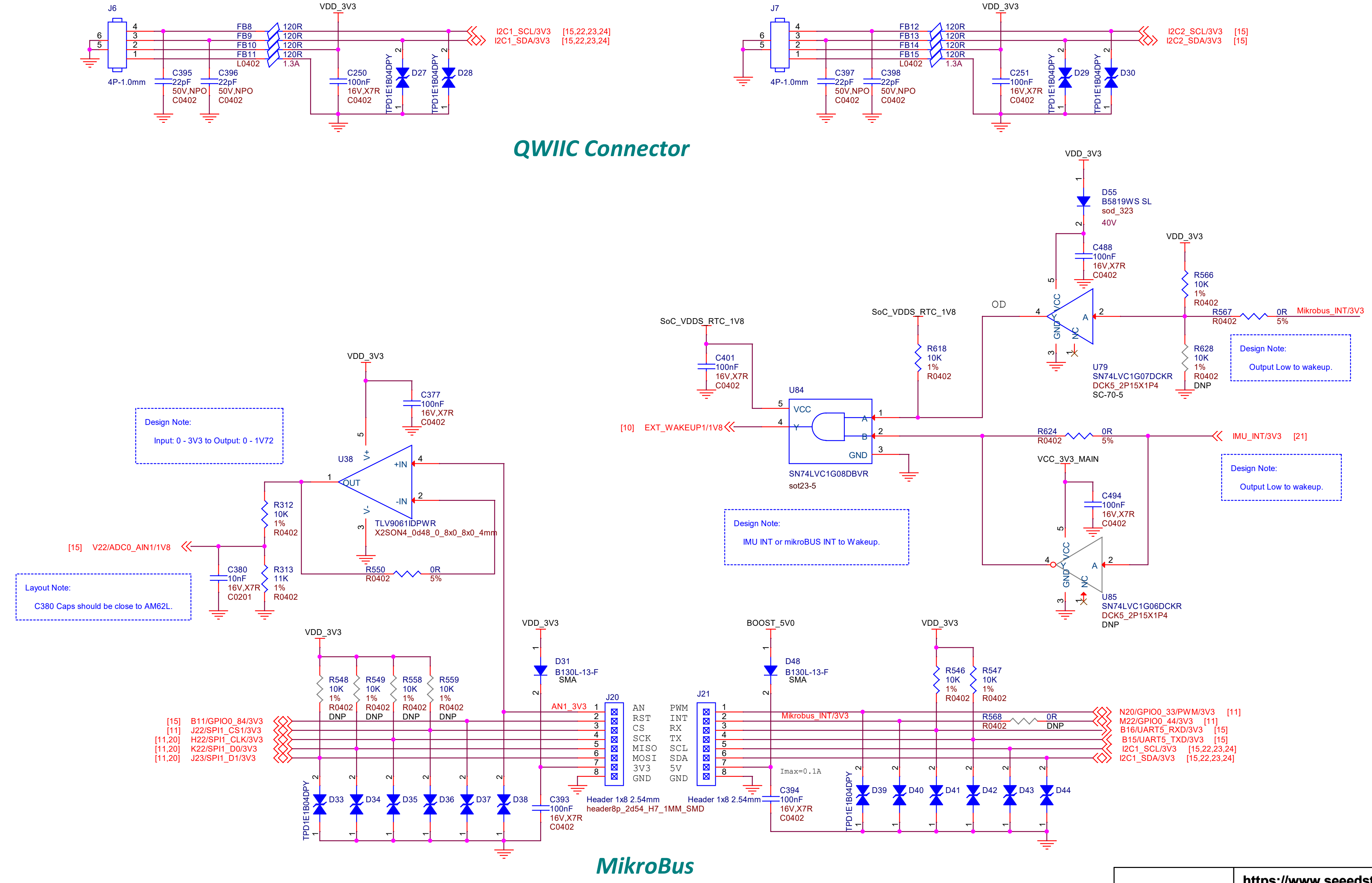

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	Title: BeagleBadge AM62L
Size: A3	Document Number: 22 QWIIC & MikroBus — Rev: RevA
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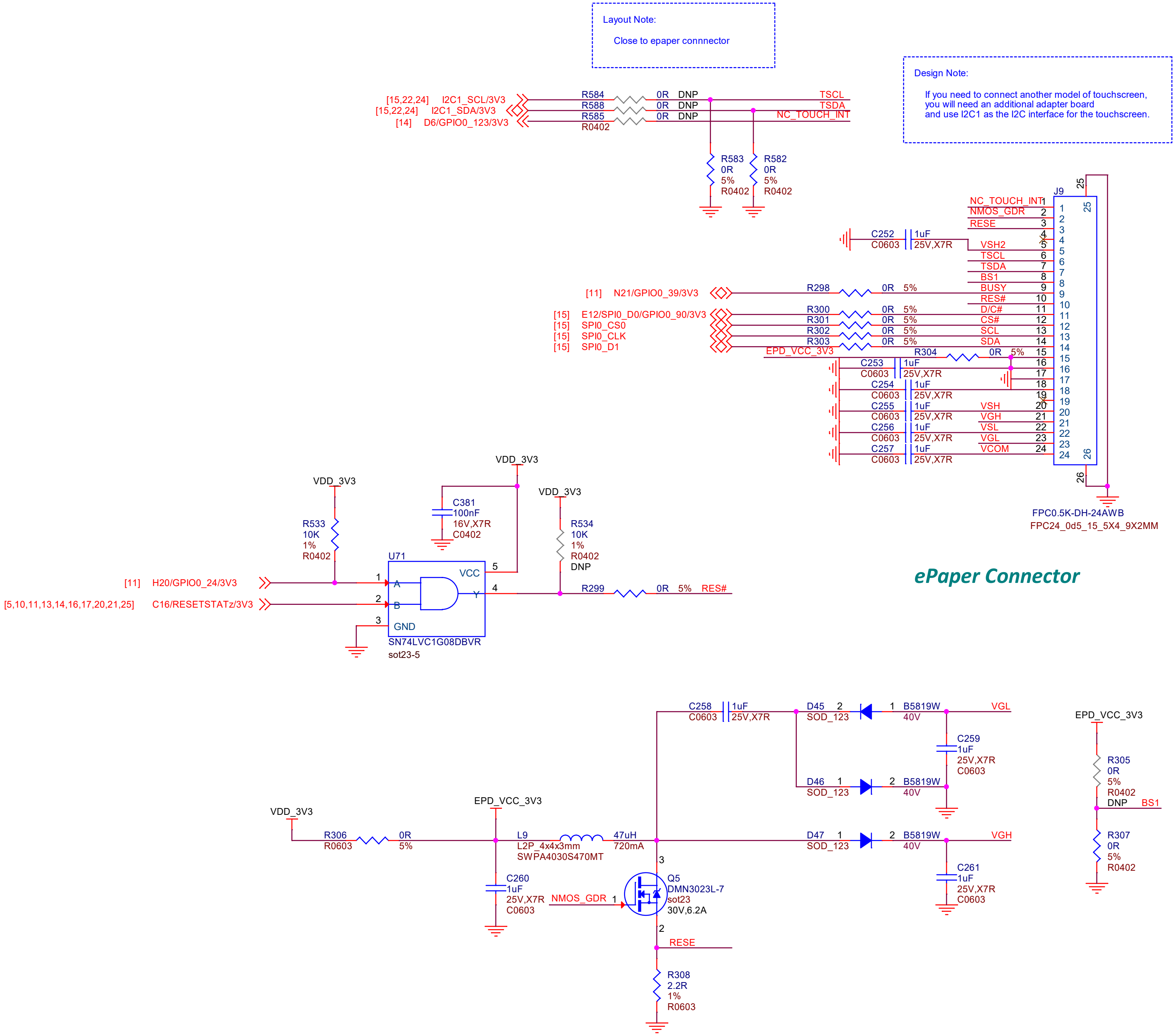

nput/Output Terminals

Pin #	Single	
1	NC	No
2	GDR	
3	RESE	
4	NC	N
5	VSH2	
6	TSCL	I2
7	TSDA	I2
8	BS1	
9	BUSY	
10	RES #	
11	D/C #	
12	CS #	
13	SCL	
14	SDA	
15	VDDIO	
16	VCI	
17	VSS	
18	VDD	
19	VPP	
20	VSH1	
21	VGH	P
22	VSL	
23	VGL	P
24	VCOM	

Design Note:

BS1 = 0, 4-Line SPI Mode
BS1 = 1, 3-Line SPI Mode

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Title: BeagleBadge AM62L

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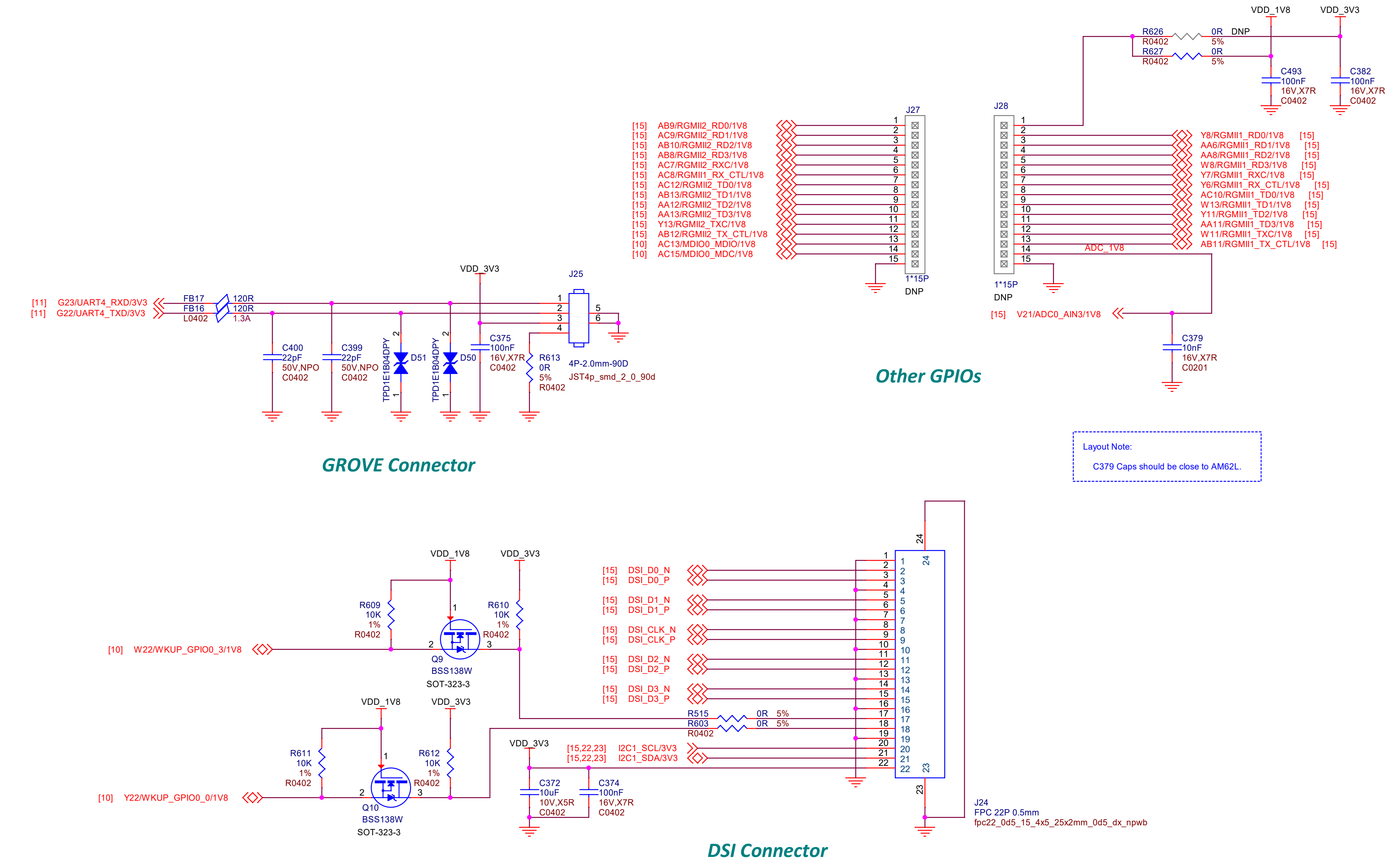

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Title: BeagleBadge AM62L		
Size: A3	Document Number: 24 Other PINs & DSI & Grove	Rev: RevA
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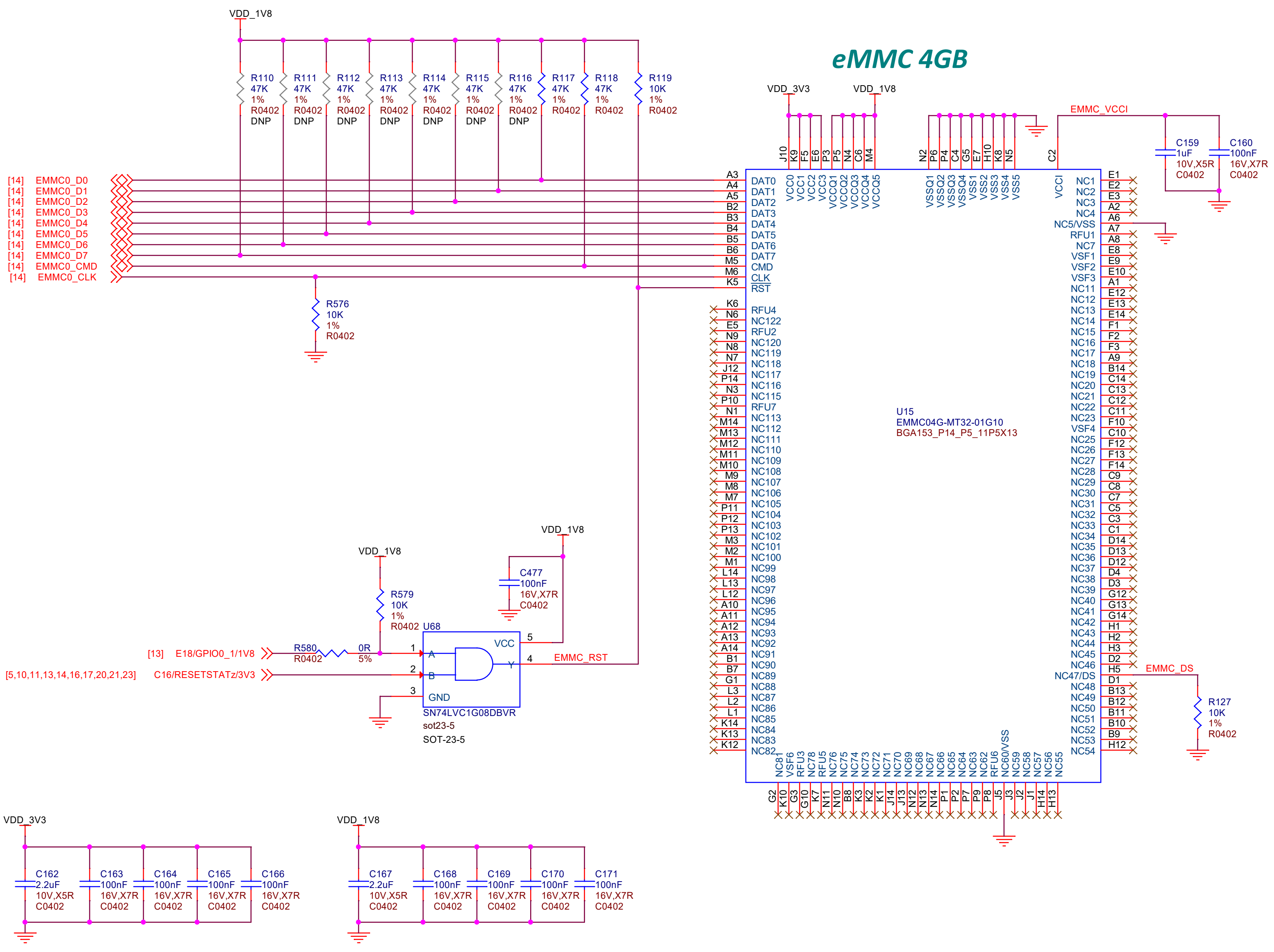

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Size:	Document Number:	Rev:
A3	25 AM62L eMMC	RevA

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